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StreamFlix Data Quality Report

Phase 1: Data Loading, Cleaning & Quality Check

Kindly view the Python Notebook file at this link

Summary

All 6 StreamFlix files were loaded and checked. The data is in good shape. There are no duplicate rows, no broken links between tables, and no sessions that run longer than the content. The only missing values are in two columns. All 8 date columns were converted from text to proper dates.

1. Dataset Dimension Summary

Dataset	Rows	Columns
subscribers	15,000	14
titles	9,000	21
watch_history	650,000	10
ratings	130,000	5
reviews	110,000	7
watchlist	65,000	5

Finding: All 6 files loaded without errors. The row counts match the sizes expected for this project. *watch_history* is the biggest table with 650,000 rows.

2. Date Column Conversion Summary

Before conversion, all date columns were read as text (object). They were converted to datetime.

Dataset	Column	Before	After	Is Date?
subscribers	signup_date	object (text)	datetime64[ns]	Yes
subscribers	churn_date	object (text)	datetime64[ns]	Yes
titles	date_added	object (text)	datetime64[ns]	Yes
titles	license_expiry	object (text)	datetime64[ns]	Yes
watch_history	watch_date	object (text)	datetime64[ns]	Yes
ratings	rating_date	object (text)	datetime64[ns]	Yes
reviews	review_date	object (text)	datetime64[ns]	Yes
watchlist	added_date	object (text)	datetime64[ns]	Yes

Finding: All 8 date columns across the 6 tables are now datetime64[ns]. Other columns already had suitable types: numbers (int64, float64) for ages, prices, durations, and scores, and bool for *is_active*, *completed*, *watched,* and *is_original*.

3. Null Value Analysis Summary

Dataset	Column	Missing Count	Null %
subscribers	churn_date	11,199	74.66
titles	license_expiry	1,966	21.84
watch_history	None	0	0.00
ratings	None	0	0.00
reviews	None	0	0.00
watchlist	None	0	0.00

Finding: Only 2 columns have missing values: *churn_date* in subscribers (74.66%) and *license_expiry* in titles (21.84%). The other 4 tables have no missing values.

- *churn_date*: 11,199 of 15,000 subscribers have no churn date.
- *license_expiry*: 1,966 of 9,000 titles have no expiry date.

4. Checking for Duplicates

Check	Duplicates Found
Duplicate rows in subscribers	0
Duplicate rows in titles	0
Duplicate rows in watch_history	0
Duplicate rows in watchlist	0
Duplicate subscriber_id in subscribers	0
Duplicate title_id in titles	0

Finding: No duplicates were found. Every subscriber_id and title_id is unique.

5. Outlier Detection: watch_duration_min

Check: Is any session watched for longer than the content itself (watch_duration_min greater than content_duration_min)?

Measure	Result
Sessions longer than content	0
Percentage of sessions flagged	0.0000%
Total sessions checked	650,000

Finding: No outliers of this type. All sessions are shorter than or equal to the content length.

6. Referential Integrity

Check: Do the subscriber_id and title_id values in watch_history exist in their parent tables?

Check	Orphaned IDs
subscriber_id in watch_history not found in subscribers	0
title_id in watch_history not found in titles	0

Finding: All links are valid. Every session in *watch_history* belongs to a real subscriber and a real title.

7. Specific Checks

Check	Result	Status
Active subscribers with a churn_date	0	Passed
Churned subscribers with churn_date on or before signup_date	0	Passed
completion_pct different from watch ÷ content × 100 by more than 1%	0 of 650,000	Passed
Reviews with sentiment other than Positive / Neutral / Negative	0	Passed
Watchlist entries with unknown subscriber_id	0	Passed
Watchlist entries with unknown title_id	0	Passed
Active subscribers with tenure of 1 month or less	321 (2.14%)	Noted

Finding: All checks passed. Churn dates are correct and match the active status. *completion_pct* matches the watch time. Sentiment has only the 3 allowed values. Every watchlist entry links to a real subscriber and title. Short tenure: 321 active subscribers (2.14%) joined 1 month ago or less. This is normal for new sign-ups and is only noted, as per the instruction.

Missing license_expiry by title type:

Type	Has expiry	Missing	Missing %
Movie	4,131	1,205	22.58
TV Show	2,903	761	20.77

Finding: Missing license_expiry is spread across both types and across all 16 genres. The missing share ranges from about 19% (Romance) to 30% (Musical) across genres. No single type or genre explains the blanks.

Overall Conclusion

The data is clean and ready for Phase 2 (EDA).
No rows were removed. No duplicates, broken links, or impossible durations were found.
Missing values in churn_date and license_expiry should be kept in mind in later analysis.